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CS 488- Homework 1

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Case Studies and Exercises by Diana Franklin

Case Study 1: Chip Fabrication Cost

Concepts illustrated by this case study

- Fabrication Cost
- Fabrication Yield
- Defect Tolerance Through Redundancy

Many factors are involved in the price of a computer chip. Intel is spending \$7 billion to complete its Fab 42 fabrication facility for 7 nm technology. In this case study, we explore a hypothetical company in the same situation and how different design decisions involving fabrication technology, area, and redundancy affect the cost of chips.

- 1.1 [10/10] <1.6> Figure 1.26 gives hypothetical relevant chip statistics that influence the cost of several current chips. In the next few exercises, you will be exploring the effect of different possible design decisions for the Intel chips.

Chip	Die Size (mm ²)	Estimated defect rate (per cm ²)	<i>N</i>	Manufacturing size (nm)	Transistors (billion)	Cores
BlueDragon	180	0.03	12	10	7.5	4
RedDragon	120	0.04	14	7	7.5	4
Phoenix ⁸	200	0.04	14	7	12	8

Figure 1.26 Manufacturing cost factors for several hypothetical current and future processors.

- a. [10] <1.6> What is the yield for the Phoenix chip?
- b. [10] <1.6> Why does Phoenix have a higher defect rate than BlueDragon?

1.1. a. Yield: $1/(1+0.04)^{14} = 0.34$

b. Phoenix has a higher defect rate than BlueDragon because it has a larger Die Size, which are more likely to have defects due to the larger size.

- 1.2 [20/20/20/20] <1.6> They will sell a range of chips from that factory, and they need to decide how much capacity to dedicate to each chip. Imagine that they will sell two chips. Phoenix is a completely new architecture designed with 7 nm technology in mind, whereas RedDragon is the same architecture as their 10 nm BlueDragon. Imagine that RedDragon will make a profit of \$15 per defect-free chip. Phoenix will make a profit of \$30 per defect-free chip. Each wafer has a 450 mm diameter.
- [20] <1.6> How much profit do you make on each wafer of Phoenix chips?
 - [20] <1.6> How much profit do you make on each wafer of RedDragon chips?
 - [20] <1.6> If your demand is 50,000 RedDragon chips per month and 25,000 Phoenix chips per month, and your facility can fabricate 70 wafers a month, how many wafers should you make of each chip?

450 mm = 45 cm

1.2 a. $\frac{((\frac{45}{2})^2 \pi)}{1.2} - \frac{45\pi}{\sqrt{2.2}} = 705.22 - 70.69 = 724.53 \approx 724$ Rounds down since we cannot exceed

$\frac{1}{(1+(0.04 \cdot 2))^{14}} = \frac{1}{2.937} = 0.340$ $724 \cdot 0.34 \cdot 30 = 7,384.80$

Phoenix Chip Profit = \$7,384.80

b. $\frac{((\frac{45}{2})^2 \pi)}{1.2} - \frac{45\pi}{\sqrt{1.2}} = 1325.36 - 91.26 = 1,234.1 \approx 1,234$

$\frac{1}{(1+(0.04 \cdot 1.2))^{14}} = \frac{1}{1.928} = 0.52$ $1234 \cdot 0.52 \cdot 15 = 9,606.69$

Red Dragon Chip Profit = \$9,606.69

c. $\frac{50000}{1234} = 40.5 \approx 40$ $\frac{25000}{724} = 34.5 \approx 34$

Since we are limited to 70 chips a month, we would make 30 Phoenix Chips and 40 Red Dragon Chips.

- 1.3 [20/20] <1.6> Your colleague at AMD suggests that, since the yield is so poor, you might make chips more cheaply if you released multiple versions of the same chip, just with different numbers of cores. For example, you could sell Phoenix⁸, Phoenix⁴, Phoenix², and Phoenix¹, which contain 8, 4, 2, and 1 cores on each chip, respectively. If all eight cores are defect-free, then it is sold as Phoenix⁸. Chips with four to seven defect-free cores are sold as Phoenix⁴, and those with two or three defect-free cores are sold as Phoenix². For simplification, calculate the yield for a single core as the yield for a chip that is 1/8 the area of the original Phoenix chip. Then view that yield as an independent probability of a single core being defect free. Calculate the yield for each configuration as the probability of at the corresponding number of cores being defect free.
- [20] <1.6> What is the yield for a single core being defect free as well as the yield for Phoenix⁴, Phoenix² and Phoenix¹?
 - [5] <1.6> Using your results from part a, determine which chips you think it would be worthwhile to package and sell, and why.

Handwritten calculations on lined paper:

1.3.a. Phoenix⁴: 4-7 defect free
 Phoenix²: 2-3 defect free
 Phoenix¹: 1 defect free

$$\sum_{i=1}^N \text{combinations}_i (0.13)^{8-N} \cdot (0.87)^N$$

$$\frac{1}{(1 + 6 \cdot 0.04 + 0.25)^{14}} = 0.87$$

Phoenix⁴: $(0.61 + 0.06 + 0.21 + 0.39) = 0.57$
 Phoenix²: $(0.0001 + 0.001) = 0.0011$
 Phoenix¹: 0.000004

- b. Based on the results, it would be worthwhile to sell Phoenix(4) as it would give the higher yield rate than Phoenix(2) and Phoenix(1).

Case Study 2: Power Consumption in Computer Systems

Concepts illustrated by this case study

- Amdahl's Law
- Redundancy
- MTTF
- Power Consumption

Power consumption in modern systems is dependent on a variety of factors, including the chip clock frequency, efficiency, and voltage. The following exercises explore the impact on power and energy that different design decisions and use scenarios have.

- 1.4 [10/10/10/10] <1.5> A cell phone performs very different tasks, including streaming music, streaming video, and reading email. These tasks perform very different computing tasks. Battery life and overheating are two common problems for cell phones, so reducing power and energy consumption are critical. In this problem, we consider what to do when the user is not using the phone to its full computing capacity. For these problems, we will evaluate an unrealistic scenario in which the cell phone has no specialized processing units. Instead, it has a quad-core, general-purpose processing unit. Each core uses 0.5 W at full use. For email-related tasks, the quad-core is $8\times$ as fast as necessary.
- a. [10] <1.5> How much dynamic energy and power are required compared to running at full power? First, suppose that the quad-core operates for $1/8$ of the time and is idle for the rest of the time. That is, the clock is disabled for $7/8$ of the time, with no leakage occurring during that time. Compare total dynamic energy as well as dynamic power while the core is running.
 - b. [10] <1.5> How much dynamic energy and power are required using frequency and voltage scaling? Assume frequency and voltage are both reduced to $1/8$ the entire time.
 - c. [10] <1.6, 1.9> Now assume the voltage may not decrease below 50% of the original voltage. This voltage is referred to as the *voltage floor*, and any voltage lower than that will lose the state. Therefore, while the frequency can keep decreasing, the voltage cannot. What are the dynamic energy and power savings in this case?

a. Dynamic Energy: $1/8$; Power: 0.5 W Each (2 W Total)

Handwritten calculations for problem 1.4a and 1.4c:

b. Given: $1/8$ Energy: $\frac{(1/8 \cdot 1/8)^2}{(1/8)^2} = \frac{(1/64)^2}{1/64} = 0.015625$

Power: $0.0156 \cdot \frac{(1/8 \cdot 1/8)}{1/8} = 0.00195$

c. $(0.5)^2 = 0.25$ Energy: 0.25

Power: 0.03125

- 1.5 [10/10/10] <1.5> As mentioned in Exercise 1.4, cell phones run a wide variety of applications. We'll make the same assumptions for this exercise as the previous one, that it is 0.5 W per core and that a quad core runs email 3× as fast.
- [10] <1.5> Imagine that 80% of the code is parallelizable. By how much would the frequency and voltage on a single core need to be increased in order to execute at the same speed as the four-way parallelized code?
 - [10] <1.5> What is the reduction in dynamic energy from using frequency and voltage scaling in part a?
 - [10] <1.5> How much energy is used with a dark silicon approach? In this approach, all hardware units are power gated, allowing them to turn off entirely (causing no leakage). Specialized ASICs are provided that perform the same computation for 20% of the power as the general-purpose processor. Imagine that each core is power gated. The video game requires two ASICs and two cores. How much dynamic energy does it require compared to the baseline of parallelized on four cores?

$$1.5. a. \frac{0.8}{4} + 0.2 = \frac{1}{0.4} = 2.5$$

$$b. 4 \text{ cores, } 4 \cdot \left(\frac{(\frac{1}{8} \cdot \frac{1}{2.5})^2}{(\frac{1}{8})^2} \right) = 0.16 \text{ kEnergy}$$

$$0.16 \left(\frac{\frac{1}{8} \cdot \frac{1}{2.5}}{\frac{1}{8}} \right) = 0.256 \text{ kPower}$$

$$c. \frac{2 + (0.2 \cdot 2)}{4} = 0.6 \text{ kDynamic Energy}$$

Exercises

- 1.7 [10/15/15/10/10] <1.4, 1.5> One challenge for architects is that the design created today will require several years of implementation, verification, and testing before appearing on the market. This means that the architect must project what the technology will be like several years in advance. Sometimes, this is difficult to do.
- [10] <1.4> According to the trend in device scaling historically observed by Moore's Law, the number of transistors on a chip in 2025 should be how many times the number in 2015?
 - [15] <1.5> The increase in performance once mirrored this trend. Had performance continued to climb at the same rate as in the 1990s, approximately what performance would chips have over the VAX-11/780 in 2025?

1.7.a. Moore's Law: Doubles every 2 years

$$2^{\frac{10}{2}} = 2^5 = 32 \text{ times}$$

b. $2025 - 1978 = 47$ $47 \cdot \frac{12}{18} = 31.33 \approx 31$

$$2^{31} = 2,147,483,648 \text{ times}$$

- 1.10 [10/10/20] <1.7> Availability is the most important consideration for designing servers, followed closely by scalability and throughput.
- [10] <1.7> We have a single processor with a failure in time (FIT) of 100. What is the mean time to failure (MTTF) for this system?
 - [10] <1.7> If it takes one day to get the system running again, what is the availability of the system?

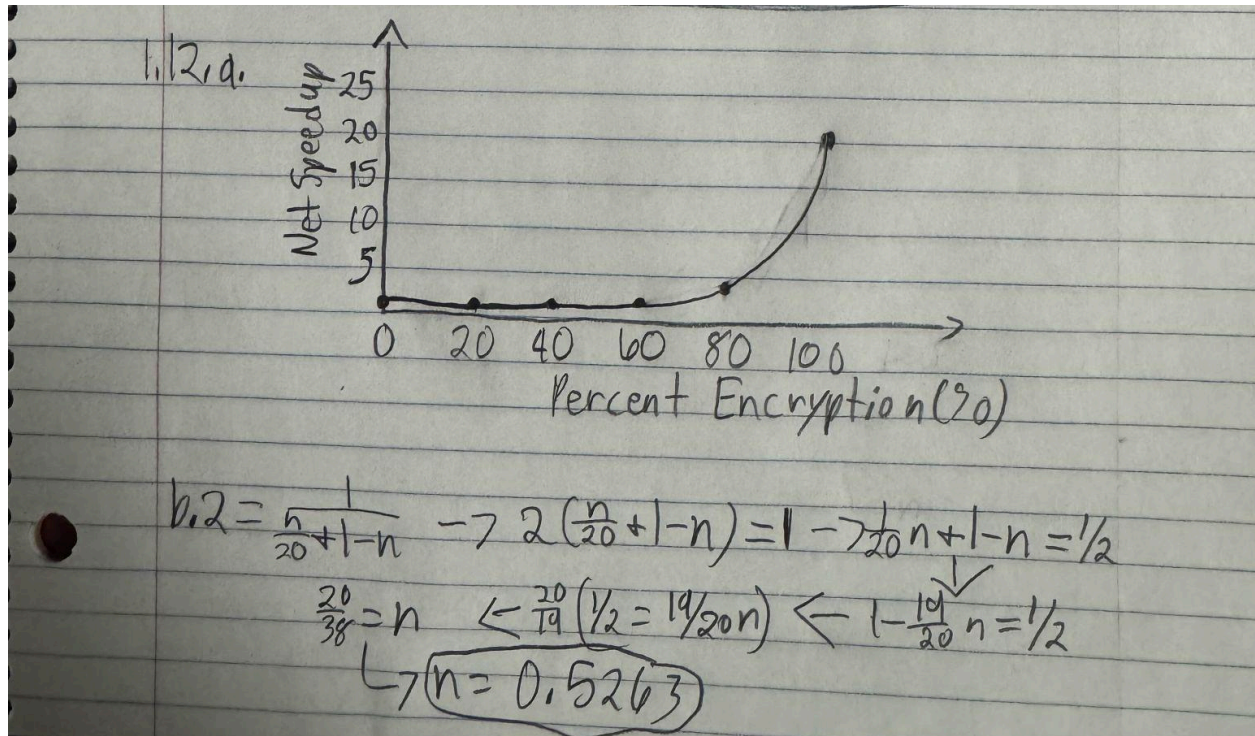
1.10.d. 100 FIT = 100 failures in 10^9 hours

$$\text{MTTF} = \frac{10^9}{100} = 10^7 \text{ hours}$$

b. $\text{Availability} = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}} = \frac{10^7}{10^7 + 24} \approx 99.9997690$

1.12 [20/10/10/10/15] <1.9> In this exercise, assume that we are considering enhancing a quad-core machine by adding encryption hardware to it. When computing encryption operations, it is 20 times faster than the normal mode of execution. We will define percentage of encryption as the percentage of time in the original execution that is spent performing encryption operations. The specialized hardware increases power consumption by 2%.

- [20] <1.9> Draw a graph that plots the speedup as a percentage of the computation spent performing encryption. Label the y-axis "Net speedup" and label the x-axis "Percent encryption."
- [10] <1.9> With what percentage of encryption will adding encryption hardware result in a speedup of 2?
- [10] <1.9> What percentage of time in the new execution will be spent on encryption operations if a speedup of 2 is achieved?



c.

$$\frac{\frac{0.5263}{20}}{(1 - 0.5263) + \frac{0.5263}{20}} = \frac{0.0263}{0.5003} = 0.0526$$